MOS Capacitor Simulation project

1 Equations

$$J_n = q\mu_n nE + qD_n \frac{dn}{dx} \tag{1}$$

$$V_t = \frac{KT}{q} \tag{2}$$

$$\phi = \frac{V}{V_t} \tag{3}$$

$$E = -\frac{dV}{dx} \tag{4}$$

$$\implies J_n = KT\mu_n \left(\frac{nE}{V_t} + \frac{dn}{dx} \right) = KT\mu_n \left(\frac{-nd\phi}{dx} + \frac{dn}{dx} \right) \tag{5}$$

$$\text{Carrier continuity equation:} \frac{\partial n}{\partial t} = \frac{1}{q}\frac{\partial J_n}{\partial x} + G_n - R_n \tag{6}$$

$$= \frac{1}{q}\frac{\partial J_n}{\partial x} - \frac{n - n_o}{\tau_n} \tag{7}$$

Before proceeding forward, it is necessary to realize that the current is essentially a quantity that is defined as a flux that goes from one point to another. Therefore, it is defined between two nodes, as the flux that enters the node. Defining current at a particular point does not carry significant meaning. So, now that we have understood the meaning of current, we define it as the flux flowing between node $i-1$ and node i. In order to uniquely represent the flux between these nodes, we will use the index $i - \frac{1}{2}$ to denote the element between nodes $i-1$ and i. Similarly, to represent the element between node i and node $i+1$, we will use the index $i + \frac{1}{2}$. Let us assume $n(x) = u(x)\, e^{\phi(x)}$. This is because the carrier concentration depends exponentially on

the potential. Also we assume potential to be linear between any two nodes. So,

$$\phi = \phi_{i-1} + \frac{\phi_i - \phi_{i-1}}{x_i - x_{i-1}}(x - x_{i-1}) = \phi_{i-1} + \frac{\phi_i - \phi_{i-1}}{\Delta x_{i-1}}(x - x_{i-1}) \tag{8}$$

$$\implies J_{i-\frac{1}{2}} = kT\mu_n e^{\phi}\frac{du}{dx} \tag{9}$$

$$\implies \int_{x_{i-1}}^{x_i} J_{i-\frac{1}{2}} e^{-\phi}\, dx = \int_{u_{i-1}}^{u_i} kT\mu_n\, du \tag{10}$$

$$\implies J_{i-\frac{1}{2}} = \frac{KT\mu_n}{\Delta x_{i-1}}\left(\frac{n_i - n_{i-1}e^{(\phi_i - \phi_{i-1})}}{e^{\phi_i - \phi_{i-1}} - 1}\right)(\phi_i - \phi_{i-1}) \tag{11}$$

$$\text{Bernoulli function: } B(x) = \frac{x}{e^x - 1} \tag{12}$$

$$\implies J_{i-\frac{1}{2}} = \frac{KT\mu_n}{\Delta x_{i-1}} B(\phi_i - \phi_{i-1})\left(n_i - n_{i-1}e^{(\phi_i - \phi_{i-1})}\right) \tag{13}$$

$$J_{i+\frac{1}{2}} = \frac{KT\mu_n}{\Delta x_i} B(\phi_{i+1} - \phi_i)\left(n_{i+1} - n_i e^{(\phi_{i+1} - \phi_i)}\right) \tag{14}$$

Since, in DC bias, steady state we have $\frac{1}{q}\frac{dJ}{dx} = \frac{n - n_{eq}}{\tau_n}$ and assuming $\Delta x_i = x_{i+1} - x_i = ab^i$. So,

$$\Delta x_i \frac{1}{q}(J_{i+\frac{1}{2}} - J_{i-\frac{1}{2}}) = \Delta x_i \Delta x_{i-1/2}\frac{n_i - n_{eq,i}}{\tau_n} \tag{15}$$

$$\implies bB(\phi_i - \phi_{i-1})\left(n_i - n_{i-1}e^{(\phi_i - \phi_{i-1})}\right) - B(\phi_{i+1} - \phi_i)\left(n_{i+1} - n_i e^{(\phi_{i+1} - \phi_i)}\right) = \Delta x_i \Delta x_{i-1/2}\frac{n_i - n_{eq,i}}{V_t \mu_n \tau_n} \tag{16}$$

$$\implies \alpha_{i-1}n_{i-1} + \alpha_i n_i + \alpha_{i+1}n_{i+1} = \frac{n_{eq,i}}{V_t\mu_n\tau_n}\Delta x_i \Delta x_{i-1/2} \quad \forall \quad i \geq 1 \tag{17}$$

$$\alpha_{i-1} = bB(\phi_i - \phi_{i-1})e^{(\phi_i - \phi_{i-1})} \tag{18}$$

$$\alpha_i = \frac{\Delta x_i \Delta x_{i-1/2}}{V_t\mu_n\tau_n} - \left(bB(\phi_i - \phi_{i-1}) + B(\phi_{i+1} - \phi_i)e^{(\phi_{i+1} - \phi_i)}\right) \tag{19}$$

$$\alpha_{i+1} = B(\phi_{i+1} - \phi_i) \tag{20}$$

Similarly for p, assume $p(x) = u(x)e^{-\phi(x)}$, so we get

$$\beta_{i-1}p_{i-1} + \beta_i p_i + \beta_{i+1}p_{i+1} = \frac{p_{eq,i}}{V_t\mu_p\tau_p}\Delta x_i \Delta x_{i-1/2} \quad \forall \quad i \geq 1 \tag{21}$$

$$\beta_{i-1} = bB(\phi_i - \phi_{i-1}) \tag{22}$$

$$\beta_i = \frac{\Delta x_i \Delta x_{i-1/2}}{V_t\mu_p\tau_p} - \left(B(\phi_{i+1} - \phi_i) + bB(\phi_i - \phi_{i-1})e^{(\phi_i - \phi_{i-1})}\right) \tag{23}$$

$$\beta_{i+1} = B(\phi_{i+1} - \phi_i)e^{(\phi_{i+1} - \phi_i)} \tag{24}$$

Boundary conditions:

$$n_1 = n_0 e^{\phi_1 - \phi_0} \tag{25}$$

$$p_1 e^{\phi_1 - \phi_0} = p_0 \tag{26}$$

$$p_{end} = p_{eq,end}, \quad n_{end} = n_{eq,end} \tag{27}$$

Now, when we apply AC signal along with DC bias,

$$V_G = V_{DC} + v e^{j\omega t} \tag{28}$$

$$\phi(x) = \phi_0(x) + \tilde{\phi}(x) e^{j\omega t} \tag{29}$$

$$n(x) = n_0(x) + \tilde{n}(x) e^{j\omega t} \tag{30}$$

$$p(x) = p_0(x) + \tilde{p}(x) e^{j\omega t} \tag{31}$$

Let $\Delta\phi_{0,i} = \phi_{0,i+1} - \phi_{0,i}$,

$$e^{\Delta\phi_{0,i} + \Delta\tilde{\phi}_i e^{j\omega t}} \sim (1 + \Delta\tilde{\phi}_i e^{j\omega t}) e^{\Delta\phi_{0,i}} \tag{32}$$

$$B(\Delta\phi_i) = B(\Delta\phi_{0,i}) + B'(\Delta\phi_{0,i}) \Delta\tilde{\phi}_i e^{j\omega t} \tag{33}$$

$$J_{i-1/2} = \frac{KT\mu_n}{\Delta x_{i-1}} \left[B(\Delta\phi_{0,i-1}) + B'(\Delta\phi_{0,i-1}) \Delta\tilde{\phi}_{i-1} e^{j\omega t} \right] \left[n_{0,i} + \tilde{n}_i e^{j\omega t} - \left(n_{0,i-1} + \tilde{n}_{i-1} e^{j\omega t} \right) e^{\Delta\phi_{0,i-1}} (1 + \Delta\tilde{\phi_{i-1}} e^{j\omega t}) \right] \tag{34}$$

Ignoring higher order perturbations in comparison to linear and constant terms,

$$J_{i-1/2} = J_{0,i-1/2} + e^{j\omega t} \frac{KT\mu_n}{\Delta x_{i-1}} \left[B(\Delta\phi_{0,i-1}) \left[\tilde{n}_i - e^{\Delta\phi_{0,i-1}} \left(\tilde{n}_{i-1} + n_{0,i-1} \Delta\tilde{\phi}_{i-1} \right) \right] + B'(\Delta\phi_{0,i-1}) \Delta\tilde{\phi}_{i-1} \left[n_{0,i} - n_{0,i-1} e^{\Delta\phi_{0,i-1}} \right] \right] \tag{35}$$

$$= J_{0,i-1/2} + \tilde{J}_{i-1/2} e^{j\omega t} \tag{36}$$

$$\tilde{J}_{i-1/2} = \frac{KT\mu_n}{\Delta x_{i-1}} \left[B(\Delta\phi_{0,i-1}) \left[\tilde{n}_i - e^{\Delta\phi_{0,i-1}} \left(\tilde{n}_{i-1} + n_{0,i-1} \Delta\tilde{\phi}_{i-1} \right) \right] + B'(\Delta\phi_{0,i-1}) \Delta\tilde{\phi}_{i-1} \left[n_{0,i} - n_{0,i-1} e^{\Delta\phi_{0,i-1}} \right] \right] \tag{37}$$

$$\tilde{J}_{i+1/2} = \frac{KT\mu_n}{\Delta x_i} \left[B(\Delta\phi_{0,i}) \left[\tilde{n}_{i+1} - e^{\Delta\phi_{0,i}} \left(\tilde{n}_i + n_{0,i} \Delta\tilde{\phi}_i \right) \right] + B'(\Delta\phi_{0,i}) \Delta\tilde{\phi}_i \left[n_{0,i+1} - n_{0,i} e^{\Delta\phi_{0,i}} \right] \right] \tag{38}$$

Now, from carrier continuity eq,

$$\frac{\partial n(x,t)}{\partial t} = \frac{1}{q} \frac{\partial J_n(x,t)}{\partial x} - \frac{n(x,t) - n_{eq}(x)}{\tau_n} \tag{39}$$

$$\frac{1}{q} \frac{\partial J_n(x,t)}{\partial x} = \frac{1}{q} \frac{J_{i+1/2} - J_{i-1/2}}{\Delta x_{i-1/2}} = \frac{V_t \mu_n}{\Delta x_{i-1/2}} \left(\alpha_{i+1} \tilde{n}_{i+1} + \alpha_i \tilde{n}_i + \alpha_{i-1} \tilde{n}_{i-1} + c_n \right) \tag{40}$$

where,

$$\alpha_{i-1} = \frac{B(\Delta\phi_{0,i-1})}{\Delta x_{i-1}} e^{\Delta\phi_{0,i-1}}, \quad \alpha_{i+1} = \frac{B(\Delta\phi_{0,i})}{\Delta x_i}$$

$$\alpha_i = -\left(\frac{B(\Delta\phi_{0,i})}{\Delta x_i} e^{\Delta\phi_{0,i}} + \frac{B(\Delta\phi_{0,i-1})}{\Delta x_{i-1}}\right)$$

$$c_n = \frac{n_{0,i} B(\Delta\phi_{0,i}) \Delta\tilde{\phi}_i e^{\Delta\phi_{0,i}} + B'(\Delta\phi_{0,i}) \Delta\tilde{\phi}_i \left(n_{0,i+1} - n_{0,i} e^{\Delta\phi_{0,i}}\right)}{\Delta x_i}$$
$$+ \frac{n_{0,i-1} B(\Delta\phi_{0,i-1}) \Delta\tilde{\phi}_{i-1} e^{\Delta\phi_{0,i-1}} + B'(\Delta\phi_{0,i-1}) \Delta\tilde{\phi}_{i-1} \left(n_{0,i} - n_{0,i-1} e^{\Delta\phi_{0,i-1}}\right)}{\Delta x_{i-1}}$$

So we get

$$\alpha_{i+1}\tilde{n}_{i+1} + \left[\alpha_i - \frac{\Delta x_{i-1/2}}{V_t \mu_n \tau_n} - \frac{j\omega \Delta x_{i-1/2}}{V_t \mu_n}\right] \tilde{n}_i + \alpha_{i-1}\tilde{n}_{i-1} + c_{n,i} = 0 \tag{41}$$

Boundary Conditions:

- At oxide,

$$0 = J_{i-1/2} = J_{i+1/2} \tag{42}$$
$$\implies B(\Delta\phi_{0,0}) \left(\tilde{n}_1 - e^{\Delta\phi_{0,0}} (\tilde{n}_0 + n_{0,0} \Delta\tilde{\phi}_0)\right) + B'(\Delta\phi_{0,0}) \Delta\tilde{\phi}_0 \left[n_{0,1} - n_{0,0} e^{\Delta\phi_{0,0}}\right] \tag{43}$$
$$\implies \alpha_1 \tilde{n}_1 + \alpha_0 \tilde{n}_0 + c_{n,0} = 0 \tag{44}$$

  where,

$$\alpha_1 = B(\Delta\phi_{0,0}) = B(\phi_{0,1} - \phi_{0,0}), \quad \alpha_0 = -B(\Delta\phi_{0,0}) e^{\Delta\phi_{0,0}}$$
$$c_{n,0} = \left[B'(\Delta\phi_{0,0}) \left(n_{0,1} - n_{0,0} e^{\Delta\phi_{0,0}}\right) - B(\Delta\phi_{0,0}) n_{0,0} e^{\Delta\phi_{0,0}}\right] \Delta\tilde{\phi}_0$$

- On the bulk side, as $\phi = 0$ in bulk, and by equilibrium/deep-bulk reservoir assumption, $\tilde{\phi} = 0$, so we get $\tilde{n}_i = 0$ in bulk boundary where we ground it.

Similarly for p, take $p = u(x) e^{\phi(x)}$, then we get

$$J_{p,i-1/2} = \frac{KT\mu_p}{\Delta x_{i-1}} B(\phi_i - \phi_{i-1}) \left(p_{i-1} - p_i e^{\phi_i - \phi_{i-1}}\right) = \frac{KT\mu_p}{\Delta x_{i-1}} B(\Delta\phi_{i-1}) \left(p_{i-1} - p_i e^{\Delta\phi_{i-1}}\right) \tag{45}$$

Under AC analysis,

$$J_{i-1/2} = \frac{KT\mu_p}{\Delta x_{i-1}} \left[B(\Delta\phi_{0,i-1}) \left[\tilde{p}_{i-1} - e^{\Delta\phi_{0,i-1}} \left(\tilde{p}_i + p_{0,i} \Delta\tilde{\phi}_{i-1}\right)\right] + B'(\Delta\phi_{0,i-1}) \Delta\tilde{\phi}_{i-1} \left[p_{0,i-1} - p_{0,i} e^{\Delta\phi_{0,i-1}}\right]\right] \tag{46}$$

$$J_{i+1/2} = \frac{KT\mu_p}{\Delta x_i} \left[B(\Delta\phi_{0,i}) \left[\tilde{p}_i - e^{\Delta\phi_{0,i}} \left(\tilde{p}_{i+1} + p_{0,i+1} \Delta\tilde{\phi}_i\right)\right] + B'(\Delta\phi_{0,i}) \Delta\tilde{\phi}_i \left[p_{0,i} - p_{0,i+1} e^{\Delta\phi_{0,i}}\right]\right] \tag{47}$$

So from carrier continuity,

$$\frac{\partial p}{\partial t} = -\frac{1}{q}\frac{\partial J_p}{\partial x} - \frac{p - p_{eq}}{\tau_p} \tag{48}$$

$$-\frac{1}{q}\frac{\partial J_p}{\partial x} = \frac{J_{i-1/2} - J_{i+1/2}}{\Delta x_{i-1/2}} = \frac{V_t \mu_p}{\Delta x_{i-1/2}} \left(\beta_{i+1}\tilde{p}_{i+1} + \beta_i \tilde{p}_i + \beta_{i-1}\tilde{p}_{i-1} + c_{p,i}\right) \tag{49}$$

where

$$\begin{aligned}
\beta_{i+1} &= \frac{B(\Delta\phi_{0,i})}{\Delta x_i} e^{\Delta\phi_{0,i}}, \quad \beta_{i-1} = \frac{B(\Delta\phi_{0,i-1})}{\Delta x_{i-1}} \\
\beta_i &= -\left[\frac{B(\Delta\phi_{0,i})}{\Delta x_i} + \frac{B(\Delta\phi_{0,i-1})e^{\Delta\phi_{0,i-1}}}{\Delta x_{i-1}}\right] \\
c_p &= \frac{-B(\Delta\phi_{0,i-1})p_{0,i}e^{\Delta\phi_{0,i-1}} + B'(\Delta\phi_{0,i-1})\left(p_{0,i-1} - p_{0,i}e^{\Delta\phi_{0,i-1}}\right)}{\Delta x_{i-1}} \Delta\tilde{\phi}_{i-1} \\
&\quad - \frac{-B(\Delta\phi_{0,i})p_{0,i+1}e^{\Delta\phi_{0,i}} + B'(\Delta\phi_{0,i})\left(p_{0,i} - p_{0,i}e^{\Delta\phi_{0,i}}\right)}{\Delta x_i} \Delta\tilde{\phi}_i
\end{aligned}$$

So we get,

$$\beta_{i+1}\tilde{p}_{i+1} + \left(\beta_i - \frac{\Delta x_{i-1/2}}{V_t \mu_p \tau_p} - \frac{j\omega \Delta x_{i-1/2}}{V_t \mu_p}\right)\tilde{p}_i + \beta_{i-1}\tilde{p}_{i-1} + c_p = 0 \tag{50}$$

Boundary Conditions:

- At oxide,

$$0 = J_{i-1/2} = J_{i+1/2} \tag{51}$$

$$\implies B(\Delta\phi_{0,0})\left(\tilde{p}_0 - e^{\Delta\phi_{0,0}}(\tilde{p}_1 + p_{0,1}\Delta\tilde{\phi}_0)\right) + B'(\Delta\phi_{0,0})\Delta\tilde{\phi}_0\left[p_{0,0} - p_{0,1}e^{\Delta\phi_{0,0}}\right] \tag{52}$$

$$\implies \beta_1\tilde{p}_1 + \beta_0\tilde{p}_0 + c_{p,0} = 0 \tag{53}$$

  where,

$$\begin{aligned}
\beta_1 &= -B(\Delta\phi_{0,0})e^{\Delta\phi_{0,0}}, \quad \beta_0 = B(\Delta\phi_{0,0}) \\
c_{p,0} &= \left[B'(\Delta\phi_{0,0})\left(p_{0,0} - p_{0,1}e^{\Delta\phi_{0,0}}\right) - B(\Delta\phi_{0,0})p_{0,1}e^{\Delta\phi_{0,0}}\right]\Delta\tilde{\phi}_0
\end{aligned}$$

- On the bulk side, as $\phi = 0$ in bulk, $\tilde{\phi} = 0$, so we get $\tilde{p}_i = 0$ in bulk boundary where we ground it.

Instead of constructing complex matrices and solving for complex $\tilde{n}, \tilde{p}$,

$$\tilde{n} = n_R + jn_I, \quad \tilde{p} = p_R + jp_I, \quad \Delta\tilde{\phi} = \Delta\phi_R + j\Delta\phi_I, \tag{54}$$

$$\alpha_{i+1}\tilde{n}_{i+1} + (a_{n,i} + jb_{n,i})\tilde{n}_i + \alpha_{i-1}\tilde{n}_{i-1} + c_{n,R} + jc_{n,I} = 0 \tag{55}$$

$$a_i = \alpha_i - \frac{\Delta x_{i-1/2}}{V_t \mu_n \tau_n}, \quad b_i = -\omega\frac{\Delta x_{i-1/2}}{V_t \mu_n} \tag{56}$$

$$\alpha_{i+1}n_{R,i+1} + a_{n,i}n_{R,i} - b_{n,i}n_{I,i} + \alpha_{i-1}n_{R,i-1} + c_{n,R,i} = 0 \quad \text{(Real part)} \tag{57}$$

$$\alpha_{i+1}n_{I,i+1} + a_{n,i}n_{I,i} + b_{n,i}n_{R,i} + \alpha_{i-1}n_{I,i-1} + c_{n,I,i} = 0 \quad \text{(Imaginary part)} \tag{58}$$